Average rates of change

1 Consider the function represented in the table:

x	-3	0	2	6
y	-4	-16	-22	-50

a Find the average rate of change between $x = -3$ and $x = 0$.

b Find the average rate of change between $x = 0$ and $x = 2$.

c Find the average rate of change between $x = 2$ and $x = 6$.

d Does the function have a constant or a variable rate of change?

2 Consider a function that passes through the following points:

$$\{(-2, -5), (1, -20), (2, -25), (7, -50), (9, -60)\}$$

Does this function have a constant or a variable rate of change?

3 Consider the function represented in the table:

x	3	6	8	13
y	-12	-15	-17	-22

a Find the average rate of change between $x = 3$ and $x = 6$.

b Find the average rate of change between $x = 6$ and $x = 8$.

c Find the average rate of change between $x = 8$ and $x = 13$.

d Do the set of points satisfy a linear or non-linear function?

4 Consider the function that passes through the following points:

$$\{(5, -13), (9, -29), (11, -37), (16, -57)\}$$

a Find the average rate of change between $(5, -13)$ and $(9, -29)$.

b Find the average rate of change between $(9, -29)$ and $(11, -37)$.

c Find the average rate of change between $(11, -37)$ and $(16, -57)$.

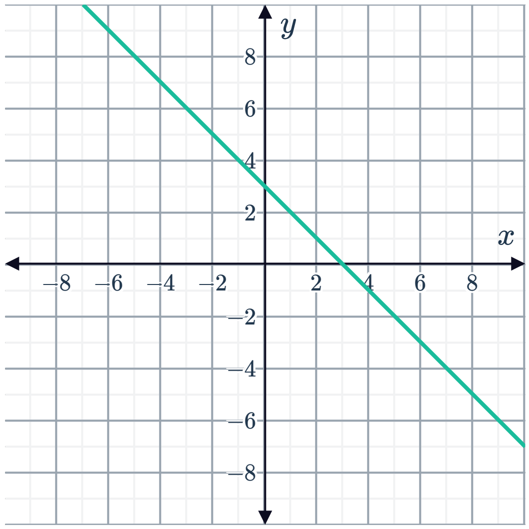
d Do the points satisfy a linear or non-linear function?

5 State whether the following functions have [redacted]stant or variable rate of change:

a $2x + y + 3 = 0$

b $y = 2^x$

c



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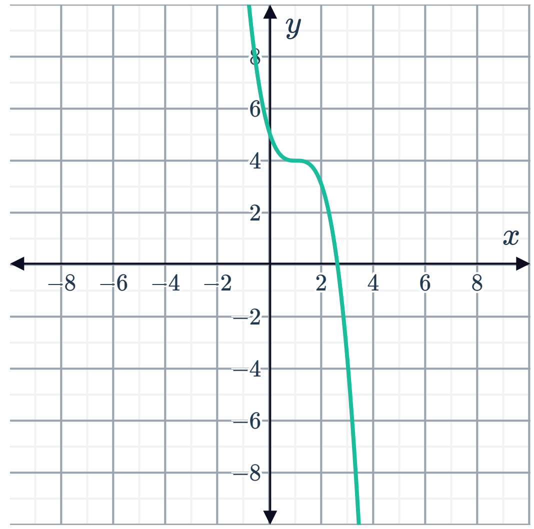

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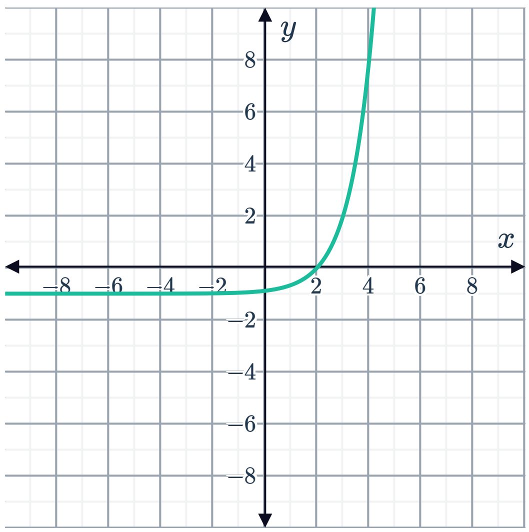

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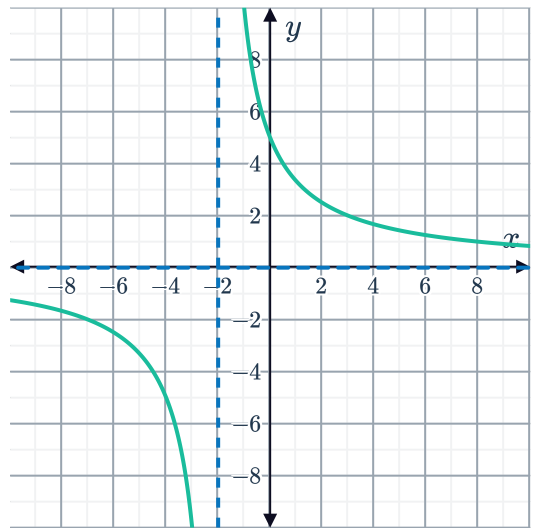

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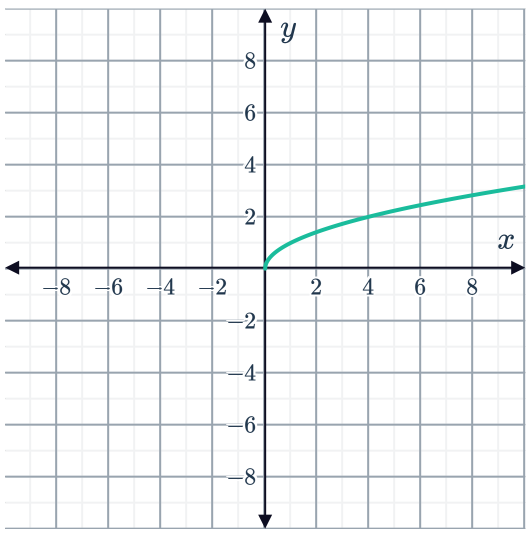

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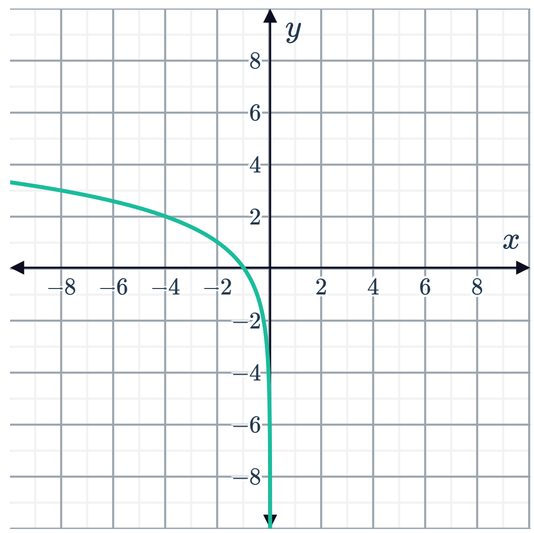


6 Consider the function that passes through the following points:

$$\{(-4,-25),(0,-2),(2,8),(7,31)\}$$

a Find the average rate of change between $(-4,-25)$ and $(0,-2)$.

b Find the average rate of change between $(0,-2)$ and $(2,8)$.

c Find the average rate of change between $(2,8)$ and $(7,31)$.

d Do the points satisfy a linear or non-linear function?

7 Consider the function represented in the table. Do the set of points satisfy a linear or non-linear function?

x	3	8	10	11	15
y	1	16	16	20	20

8 Consider the function $y = 2^x$.

a Find the average rate of change over the interval $[0,1]$.

b Find the average rate of change over the interval $[1,2]$.

c Find the average rate of change over the interval $[2,3]$.

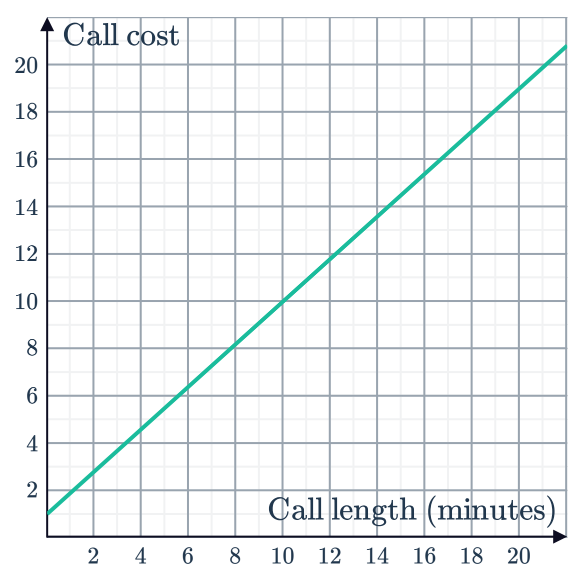
d Now, complete the following table:

Interval	$[3,4]$	$[4,5]$	$[5,6]$
Average rate of change			

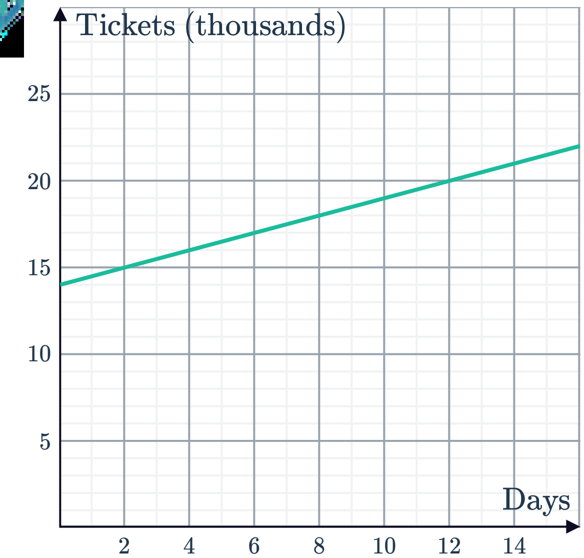
9 The graph shows the cost, in dollars, of a phone call for different call durations:

a Find the average rate of change.

b Interpret the average rate of change in the given context.

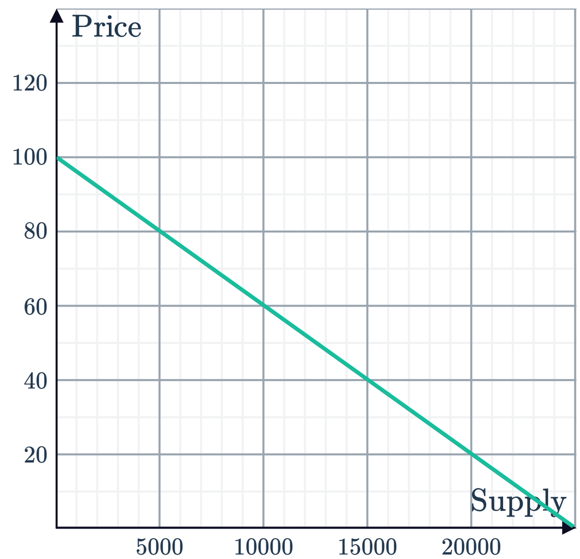


10 The graph shows the total number of tickets, in thousands, purchased for a premiering movie, with pre-ordered tickets counted as being purchased on the 0th day:



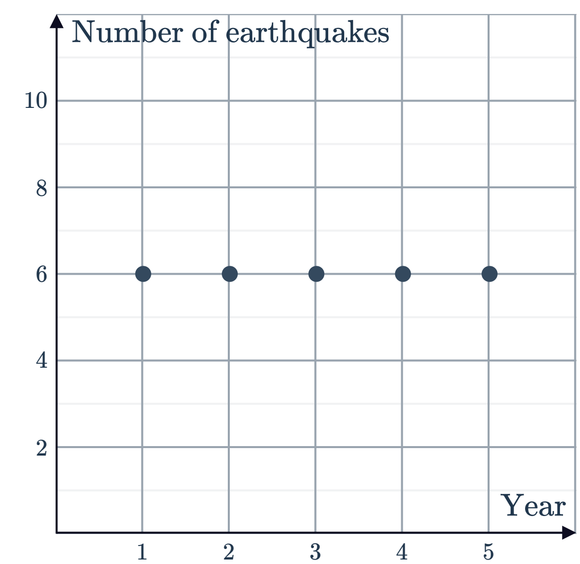
- **a** Find the average rate of change.
- **b** Interpret the average rate of change in the given context.

11 The graph shows the price, in dollars, of a new digital toy for different levels of supply:



- **a** Find the average rate of change.
- **b** Interpret the average rate of change in the given context.

12 The graph shows the number of earthquakes that a particular country experiences in each year:



- **a** Find the average rate of change.
- **b** Interpret the average rate of change in the given context.

Applications

13 The population of koalas in a particular area can be modeled by the equation $y = 10 + 2^x$, where x is the number of years from now.

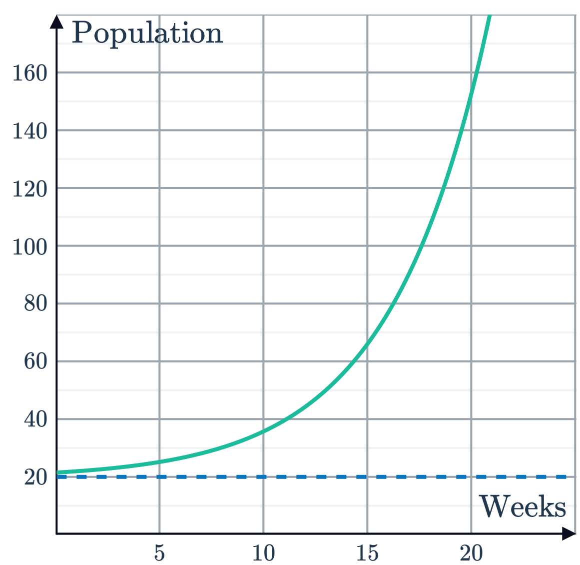
- **a** Find the value of y when $x = 0$.
- **b** Find the value of y when $x = 4$.
- **c** Now, find the population's average rate of change over the interval $[0, 4]$.

14 The daily net profit of an upmarket restaurant can be modeled by the equation $y = -17x^2 + 493x$, where x is the number of customers.
Find the average rate of change in net profit over the interval $[0, 13]$.

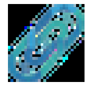
15 The value of a particular coin can be modeled by the equation $y = 80\left(2^x\right)$, where x is the number of years from now.
Find the average rate of change of its value over the interval $[0, 4]$.

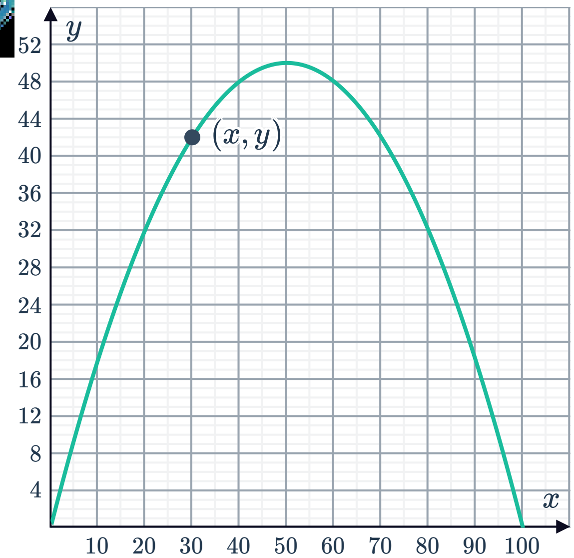
16 The path of water projected from a fountain can be modeled by the equation $y = -20x^2 + 120x$, where x is the horizontal distance from the nozzle and y is the height.
Find the average rate of change of the water's height over the interval $[0, 3]$.

17 The graph shows an insect population over a number of weeks:
Find the population's average rate of increase over the interval $[0, 20]$.



18 The side profile of a bridge is displayed on the coordinate plane such that (x, y) represents a point on the arc that is x meters horizontally from one end of the bridge and y meters vertically above the road. The bridge meets the road at coordinates $(0, 0)$ and $(100, 0)$.
Find the average steepness of the climb over the interval $[0, 20]$.

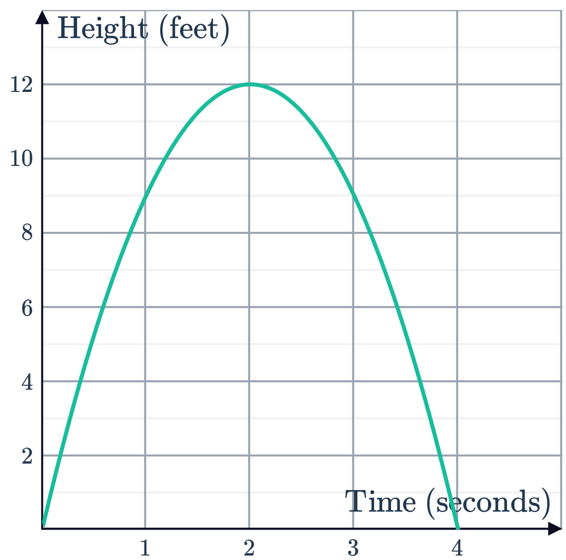




19 The graph shows the height of a cricket ball, in feet, after it is thrown:

a Find the rate of change of the height of the ball in the interval between:

- **i** When it is initially thrown and when $t = 1$.
- **ii** When $t = 1$ and when it is at its greatest point.
- **iii** When it is at its greatest point and $t = 3$.
- **iv** When $t = 3$ and when it returns to the ground.



b In which of the above intervals is the ball traveling at its fastest speed?

c What do the negative rates of change indicate?

20 Due to adverse market conditions, Tobias ar[redacted]en have had to reduce the number of staff at their respective companies, which currently have 310 staff.

- Tobias plans on reducing staff numbers by 41 each year for the next five years.
- Gwen plans on reducing staff numbers by 8 next year, 16 in the year after next, 24 in the year after that, and so on.

Year	0	1	2	3	4	5
No. Tobia's staff	310					
No. Gwen's staff	310					

a Complete the given table. Note that Year 0 represents the current year.

b What will be the rate of change in the size of Tobias's staff between years 3 and 4?

c What will be the rate of change in the size of Gwen's staff between years 3 and 4?

d Is the function representing the size of Tobias's staff over time linear or non-linear? Explain your answer.

e Is the function representing the size of Gwen's staff over time linear or non-linear? Explain your answer.

f Whose staff will decrease more quickly between years 3 and 4?

g Whose staff will decrease more quickly between years 10 and 11?

21 Mario and Christa have both recently started their own small businesses, and initially neither of them had any clients. In it's first year:

- No. Mario's client base increased by 27 per month.
- No. Christa's client base increased by 6 in its first month, 12 in its second month, 18 in its third month, and so on.

Month	0	1	2	3	4	5
Mario's clients	0					
Christa's clients	0					

a Complete the given table.

b Is the function representing the size of Mario's client base linear or non-linear?

c Is the function representing the size of Christa's client base linear or non-linear?

d Find the rate of change in the size of Mario's client base between months 2 and 3.

e Find the rate of change in the size of Christa's client base between months 2 and 3.

f Whose client base increased more quickly between months 2 and 3?

g Whose client base increased more quickly between months 4 and 5?

22 Harry and Skye each have $\$2000$ in savings [redacted]est.

- Harry chooses to put his savings in a fund that generates a return of $\$60$ each month.
- Skye chooses to put her savings in a fund that generates a return of 2% each month.

Month	0	1	2	3	4	5
Value of Harry's investment	2000					
Value of Skye's investment	2000					

a Complete the given table.

b Find the rate of change of Harry's investment between months 2 and 3.

c Find the rate of change of Skye's investment between months 2 and 3.

d Is the function representing the value of Harry's investment linear or non-linear?

e Is the function representing the value of Skye's investment linear or non-linear?

f Whose investment increased the most in value between months 2 and 3?

23 The government is concerned about the impact of a wind farm on the population of local birds and is seeking to impose a limit on the number of new turbines that may be built on the farm each year. It has proposed two options to the operators of the wind farm:

- **Option A**: The initial number of turbines is 20, and this number can be increased by 4 per year.
- **Option B**: The initial number of turbines is also 20, but the number of turbines can be increased by 10% per year.

Let y be the number of turbines, x years from now.

a Form an equation that will model the number of turbines on the farm in x years under option A.

b Form an equation that will model the number of turbines on the farm in x years under option B.

c What will be the average rate of change in the number of turbines over the next 28 years under option A?

d What will be the average rate of change in the number of turbines over the next 28 years under option B? Round your answer to two decimal places.

e Under which option will average annual growth in the number of turbines over the next 28 years be less?